Fernando Daniel García

Personal details

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Current position

Principal Professional (Research Support Staff)
Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET, *National Scientific and Technical Research Council*)
Centro de Investigaciones del Mar y la Atmósfera (CIMA, *Center for Ocean and Atmospheric Research*)
Buenos Aires - Argentina

Professional summary

Computer scientist with experience as a member of research teams in a range of scientific fields. He has developed software and designed innovative technological solutions in mathematics, ecology and, currently, the climate sciences, contributing to the transfer of knowledge to society. He has worked in artificial intelligence since 1984: symbolic artificial intelligence and expert systems (Prolog, LISP and his own inference engine), evolutionary computation (genetic algorithms) and, currently, language models and the integration of artificial intelligence into production systems.

Technical skills

Programming languages: Python, JavaScript/TypeScript, C/C++, C#, Java; Prolog and LISP (expert systems).
Web development and visualization: Node.js, React, Django, REST APIs, GraphQL; web maps (OpenLayers, MapLibre) and charts (Recharts).
Data and scientific computing: NumPy, Matplotlib; processing of satellite data (GOES-16) and forecast data (GFS); automated data collection (Playwright).
Artificial intelligence: symbolic, since 1984: expert and rule-based systems in Prolog and LISP, his own inference engine based on Robinson's resolution rule (in C, 1989) and the Augur language for expert systems; evolutionary computation: genetic algorithms (course with David E. Goldberg, Illinois, 2000; GECCO 2000 paper cited in 16 patents; applied in the 2009 game for Axe); currently, AI systems architecture design and orchestration of multiple models in production (LLMs, RAG, neural networks); Hugging Face (Transformers, sentence-transformers, PEFT/LoRA), PyTorch, TensorFlow, pgvector, OpenCV, Tesseract.
Databases and infrastructure: PostgreSQL (ORMs: Prisma, Django), Docker, GitHub.
Domain-specific languages (DSLs): design and implementation of several DSLs for different kinds of applications; currently Haiku, for full-stack applications, under continuous development.

Languages

Spanish (native), English (intermediate), Portuguese (intermediate).

Education

1983	Programador Superior (computer programming degree)	Facultad de Ciencias Físico-Matemáticas y Naturales (*School of Physical, Mathematical and Natural Sciences*) Universidad Nacional de San Luis (UNSL, *National University of San Luis*) San Luis - Argentina

Scientific and technological contributions

2022-2024	Development of a web application that integrates multiple sources of hydroclimatic information to strengthen people-centered early warning systems, as part of the IMPACT.AR program	CIMA/CONICET-UBA Ministerio de Ciencia y Tecnología de la Nación (MINCYT, *National Ministry of Science and Technology*)
2022	Development of a web application for recording severe weather events in Argentina, Uruguay, Paraguay, Chile and Brazil. PREVENIR project	Japan International Cooperation Agency (JICA) Servicio Meteorológico Nacional (SMN, *National Meteorological Service*)
2020	Development of the CIMA-GOES library for bulk download and processing of GOES-16 satellite files	CIMA/CONICET-UBA
2020	Web application for monitoring and forecasting Covid-19 in Argentina	CIMA/CONICET-UBA
2019-2024	Web application for identifying overshooting tops in satellite images, to train detection methods based on artificial intelligence.	CIMA/CONICET-UBA
2018-2019	Software development for planning the field experiment: • System that automates the download, processing and publication of GFS data • Scrapers/robots that download data from weather stations in real time	Remote sensing of Electrification, Lightning, And Mesoscale/microscale Processes with Adaptive Ground Observations (RELAMPAGO) Clouds, Aerosols, and Complex Terrain Interactions (CACTI)
2016-2023	Development of a system of scrapers/robots for collecting data from weather stations, with normalization of the collected data.	CIMA/CONICET-UBA

2015-2026	Software and visualization development for the Anticipando la Crecida project (community-based flood early warning systems), from its initial microsite to the data tools used in the co-design	CIMA/CONICET-UBA
2014-2019	Development of CIMAPS, a web platform for visualizing georeferenced information	Centro de Investigaciones del Mar y la Atmósfera (CIMA) CONICET Universidad de Buenos Aires (UBA, *University of Buenos Aires*)
2004	Development of graphing software for the characterization of sustainability syndromes in development: the case of Argentina.	Sustainable Development and Human Settlements Division Economic Commission for Latin America and the Caribbean (ECLAC) United Nations
2001-2011	Software development for research in ecology, wildlife management and epidemiology: • Population dynamics and simulation models for the sustainable management of renewable natural resources • Spatial and statistical analysis of field sampling (capybaras, the turquoise-fronted amazon parrot, wild kissing bugs, the southern green stink bug and its parasites) • Geographic distribution and species richness of the triatomine vectors of Chagas disease: climate analyses, neural networks and meta-analysis of life cycles • Computational methods for the sensitivity analysis of dynamic models • Databases of ecological literature and of environmental impact assessment; the Ecopaedia ecological glossary • Robots for downloading scientific literature, georeferenced graphing and digital image processing	Centro de Estudios Parasitológicos y de Vectores (CEPAVE, *Center for Parasitological and Vector Studies*) CONICET Universidad Nacional de La Plata (UNLP, *National University of La Plata*)
2000	Development of a website to disseminate Property Rights over Land and Ownership of Natural Resources in the Context of Environmental Sustainability in a Market Economy	Lincoln Institute of Land Policy Universidad de Belgrano (UB) Federación Argentina de Comercialización e Industrialización de la Fauna (FACIF, *Argentine Federation for Wildlife Trade and Industry*)
1995-1999	Development of an interactive web application implementing population dynamics models	Environmental Software and Services GmbH Gumpoldskirchen - Austria

1995	Development of a database of species associated with weed control	Weed Biological Control Laboratory United States Department of Agriculture
1994-1995	Development of the database for the Argentine Patagonian Coastal Zone Management Plan	Global Environment Facility (GEF) World Bank United Nations Development Programme (UNDP)
1989-1993	Software development: • Inference engine for building expert systems. • LR/LALR parser generator. • Hypertext engine for the Directory of Ecologists and Environmental Scientists of Latin America, the Caribbean and the Iberian Peninsula	Departamento de Biología (*Department of Biology*) Facultad de Ciencias Exactas y Naturales (*School of Exact and Natural Sciences*) Universidad de Buenos Aires (FCEN-UBA)
1988-1989	Research project: expert system for natural resource management	Dirección de Ecología de la Provincia de Neuquén (*Directorate of Ecology of the Province of Neuquén*)
1987	Development of a prototype system for monitoring, abnormal event annunciation and control rod interlock logic for the RA3 nuclear research reactor	Comisión Nacional de Energía Atómica (CNEA, *National Atomic Energy Commission*)
1984-1996	Software development in the following areas: • Network reliability • Graph theory • Symbolic differentiation	Instituto Argentino de Matemática (IAM, *Argentine Institute of Mathematics*) Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET)

Research projects

2022-2026	The project for numerical weather prediction and warning communication system for densely populated and vulnerable cities	Japan International Cooperation Agency (JICA)
2022-2024	Very short-range weather forecasting using radar data assimilation and machine learning	Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET)
2022-2024	People-centered early warning system in the Municipality of La Matanza, for the IMPACT.AR program.	Ministerio de Ciencia, Tecnología e Innovación (MINCYT, *Ministry of Science, Technology and Innovation*) CIMA/CONICET-UBA Municipio de La Matanza

2018-2022	RELAMPAGO: an opportunity to improve the understanding of the physical and dynamical processes that initiate and sustain high-impact weather events International field campaign (Argentina, the United States and Brazil; June 2018 – April 2019). Contributions: real-time collection of weather station data (Argentina Mesonet Data dataset, NCAR) and processing of GFS forecasts for operations planning.	Departamento de Ciencias de la Atmósfera y los Océanos (DCAO, *Department of Atmospheric and Ocean Sciences*) Facultad de Ciencias Exactas y Naturales Universidad de Buenos Aires (FCEN-UBA)
2015-2026	Anticipando la Crecida: co-design and implementation of community-based flood early warning systems in La Matanza, Lomas de Zamora, Avellaneda and Lanús (Buenos Aires Metropolitan Area). Contributions: data curation, methodology, software development and visualization.	Centro de Investigaciones del Mar y la Atmósfera (CIMA) CONICET Universidad de Buenos Aires (UBA)
1995-1999	GAIA: A Multi-Media Tool for Natural Resources Management and Environmental Education	Environmental Software and Services GmbH Gumpoldskirchen - Austria
1994-1995	PMZCP: Argentine Patagonian Coastal Zone Management Plan	Global Environment Facility (GEF) Argentina

Publications

Simone, I., Salio, P., Ruiz, J., Cancelada, M., Vidal, L. and **García, F.** (in preparation). A short climatology of overshooting cloud top detection developed from a simplified method in southern South America.

Robledo, F., Moreira, D., Prudente, C., … **García, F.**, … Pérez Vega, A. E. [36 authors] (2026). Co-design and implementation of community-based flood early warning systems in eastern Argentina. *Environmental Research: Climate, 5*(3), 035041. https://doi.org/10.1088/2752-5295/ae7fac

Salio, P., Bechis, H., Ribeiro, B. Z., de Lima Nascimento, E., Galligani, V., **García, F.**, … Villagrán Asiares, C. I. [39 authors] (2024). Toward a South American High-Impact Weather Reports Database. *Bulletin of the American Meteorological Society, 105*(7), E1204–E1217. https://doi.org/10.1175/BAMS-D-23-0063.1

García, F., Ruiz, J., Salio, P., Bechis, H. and Nesbitt, S. (2019). *Argentina Mesonet Data* (Version 1.1) [Data set]. UCAR/NCAR – Earth Observing Laboratory. https://doi.org/10.26023/JEXB-W6B6-E310. Also in CONICET's research data repository (2024): http://hdl.handle.net/11336/243011

García, F. (2000). Computer Screen Design Aided by a Genetic Algorithm. In D. Whitley (Ed.), *Late Breaking Papers at the 2000 Genetic and Evolutionary Computation Conference* (pp. 98–101).

Cited as prior art in 16 US patents from a single line of technology (2006–2023), assigned to Affinnova, Inc. and Nielsen; among them, US 7,016,882 and US 9,785,995.

Presentations at conferences and meetings

2022	"Towards a South American severe storm database". Co-author, with H. Bechis, B. Ribeiro, P. Salio and others.	30th Conference on Severe Local Storms American Meteorological Society (AMS) Santa Fe, New Mexico - USA
2022	"Hacia la creación de una base de datos de tormentas severas en Sudamérica" (*towards a severe storm database for South America*). Co-author, with H. Bechis, B. Zanetti Ribeiro, P. Salio and others. Abstract: http://hdl.handle.net/20.500.12160/2474	XIV Congreso Argentino de Meteorología (CONGREMET XIV, *14th Argentine Congress of Meteorology*) Centro Argentino de Meteorólogos Buenos Aires - Argentina
1996	"Argentina case study"	GAIA Project Startup - Meeting Environmental Software and Services GmbH Gumpoldskirchen - Austria
1991	"Sistema Experto para el Manejo del Zorro Colorado de Patagonia" (*an expert system for the management of the Patagonian red fox*)	3er Encuentro Regional de Ingeniería de Sistemas & Workshop CLEI Sistemas Expertos y Multimedios (*3rd Regional Systems Engineering Meeting and CLEI Workshop on Expert Systems and Multimedia*) Concepción - Chile

Courses and seminars taught

2001	• Application development with Delphi • Multi-tier application design with WebSnap and DataSnap	Borland Argentina Buenos Aires
1991	Introduction to Compiler Theory	Estudio Weber, Ferro & Asociados Buenos Aires
1985	Seminar on "Descriptive Logic Languages and Functional Programming"	Instituto Argentino de Matemática
1985	Co-taught the theoretical and practical course "Databases for the Management and Conservation of Wildlife and Protected Natural Areas"	Secretaría de Agricultura, Ganadería y Pesca de la Nación (*National Secretariat of Agriculture, Livestock and Fisheries*) Ministerio de Economía (*Ministry of Economy*) Buenos Aires
1984	Co-taught the graduate seminar on "Information Systems and Their Application to Urban and Regional Planning"	Facultad de Arquitectura y Urbanismo (*School of Architecture and Urban Planning*) Universidad de Buenos Aires

Courses and further training

2000	Genetic Algorithms in Search, Optimization, and Machine Learning	University of Illinois at Urbana-Champaign
1987	• Design of Language-Centered Environments • Knowledge Engineering and Expert Systems • Computerized Industrial Automation	Escuela Argentino Brasileña de Informática (EBAI, *Argentine-Brazilian School of Computer Science*) Tandil - Argentina
1986	• Applied Artificial Intelligence • Theory of Computation • Program Verification and Derivation	Escuela Argentino Brasileña de Informática (EBAI) Campinas - Brazil
1984	• Local Area Network Technology • Microprocessor Architecture • LSI: Structured Design of Integrated Circuits	Centro Latinoamericano de Matemática e Informática (CLAMI, *Latin American Center for Mathematics and Computer Science*) Buenos Aires - Argentina
1983	Introduction to Reliability	Instituto de Matemática Pura e Aplicada (IMPA, *Institute for Pure and Applied Mathematics*) Rio de Janeiro - Brazil

Fellowships

1983	Research initiation fellowship	Instituto de Matemática Pura e Aplicada (IMPA) Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq, *National Council for Scientific and Technological Development*) Rio de Janeiro - Brazil

Memberships

1996-2016	Member	Association for Computing Machinery (ACM) New York - USA
2000-2001	Member. The society, which ran the GECCO conference, became ACM SIGEVO in 2005.	International Society for Genetic and Evolutionary Computation (ISGEC) Stanford, California - USA